

Multiple Choice

1. **Answer:** A

Options: A) 2 mA.; B) 5 mA.; C) 1.5 mA.; D) 0.5 mA.; E) 6 mA.

Note: Correct option: A - 2 mA.

2. **Answer:** A

Options: A) it is a square wave signal; B) it has a constant frequency; C) it is positive for one half cycle; D) it can only have a single value at a time; E) it has a sinusoidal waveform

Note: Correct option: A - it is a square wave signal

3. **Answer:** A

Options: A) Zinc silicate.; B) Calcium sulphide.; C) Zinc oxide.; D) Zinc sulphide.; E) Calcium silicate.

Note: Correct option: A - Zinc silicate.

4. **Answer:** A

Options: A) $[(2 e^{-4s})/s^3]$; B) $2 / s^3$; C) $(2 / s^2) * e^{-4s}$; D) $4/s^3$; E) $[(2 e^{-4s})/s^4]$

Note: Correct option: A - $[(2 e^{-4s})/s^3]$

5. **Answer:** E

Options: A) 132 ohms; B) 377 ohms; C) 150 ohms; D) 300 ohms; E) 266 ohms

Note: Correct option: E - 266 ohms

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $2 / s^3$ '.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $C = (/ d)$ '.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1484.4 lb./in.'.

Long Form Response

11. **Answer:** $W = (AV_0) (1 / p_2 - 1 / p_1)$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $E = - (1/2)bB_0 e^{btra_} \text{extphi}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key